

Keano Aquino - Chrome Dev Tools Implementations

Message Logging:

- Log Info & Log Table: Updated ready block to log when the app starts and prints the initial pet statistics as a table.

```
$(function() {  
  console.info("Giga Pet application initialized."); // Log Info  
  console.table(pet_info); // Log Table  
  
  // Called function to update the name, happiness, and weight of our pet in our HTML  
  checkAndUpdatePetInfoInHtml();  
  
  // Button click events  
  $('.treat-button').click(clickedTreatButton);  
  $('.play-button').click(clickedPlayButton);  
  $('.exercise-button').click(clickedExerciseButton);  
  $('.nap-button').click(clickedNapButton); // New button event listener  
});
```

Giga Pet application initialized. [script.js:2](#)

[script.js:3](#)

(index)	Value
name	'Jiji'
weight	50
happiness	50

▼ Object [i](#)

happiness: 50

name: "Jiji"

weight: 50

▶ [\[\[Prototype\]\]](#): Object

- Log Warning & Log Error: Updated function to include warnings and errors when the pet's statistics get too low.

```
function checkWeightAndHappinessBeforeUpdating() {
  if (pet_info.weight <= 10) {
    console.warn("Warning: Jiji's weight is dangerously low!"); // Log Warning
  }

  if (pet_info.weight < 0) {
    console.error("Error: Weight cannot be less than zero. Resetting to 0."); // Log Error
    pet_info.weight = 0;
  }

  if (pet_info.happiness < 0) {
    console.error("Error: Happiness cannot be less than zero. Resetting to 0."); // Log Error
    pet_info.happiness = 0;
  }
}
```

2 ▶ Warning: Jiji's weight is dangerously low! [script.js:105](#)

✖ ▶ Error: Weight cannot be less than zero. Resetting to 0. [script.js:109](#)

- Log Group & Log Custom: Updated clickedTreatButton to group logs and use custom CSS

```
function clickedTreatButton() {
  console.group("Treat Button Clicked"); // Log Group (Start)

  pet_info.happiness += 5;
  pet_info.weight += 2;

  // Log Custom (CSS styling)
  console.log("%c Jiji was fed a treat! Stats increased.", "color: #ff9900; font-weight: bold; font-size: 1.2em;");

  triggerPetReaction(imgHappy, "Yum! That was delicious!");
  checkAndUpdatePetInfoInHtml();

  console.groupEnd(); // Log Group (End)
}
```

▼ Treat Button Clicked [script.js:29](#)

[script.js:35](#)

Jiji was fed a treat! Stats increased.

View Messages logged by the browser

- Cause 404 Network Error: Added a “fake” image element in index.html to trigger a file not found error

```

```

✖ GET [index.html:21](#) ↕

<http://127.0.0.1:5500/images/this-image-does-not-exist.png>

404 (Not Found)

- Cause Type Error: Updated clickedExerciseButton function to call a function that does not exist

```
function clickedExerciseButton() {  
  pet_info.happiness -= 10;  
  pet_info.weight -= 10;  
  
  // Cause TypeError intentionally  
  let uninitializedVariable;  
  uninitializedVariable.doSomething();  
  
  triggerPetReaction(imgSad, "Phew... I'm exhausted.");  
  checkAndUpdatePetInfoInHtml();  
}
```

```
✖ ▶ Uncaught TypeError: Cannot read properties of undefined (reading 'doSomething')  
    at HTMLButtonElement.clickedExerciseButton (script.js:56:25)  
    at HTMLButtonElement.dispatch (jquery-2.2.1.min.js:3:7516)  
    at r.handle (jquery-2.2.1.min.js:3:5597)
```

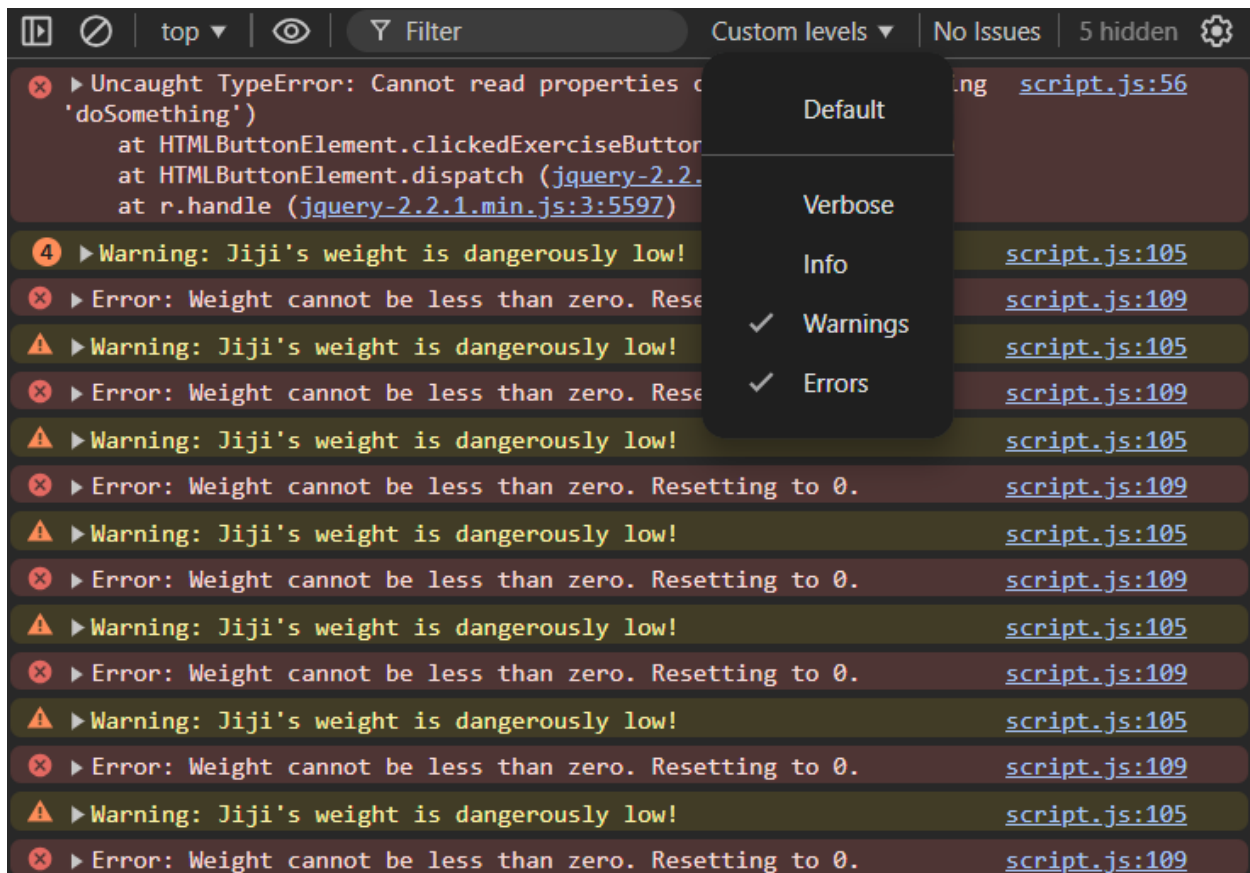
- Cause a Violation: Added 3 second delay for clickedNapButton function to trigger a violation

```
function clickedNapButton() {  
  // Cause Violation: Blocking the main thread intentionally for 3 seconds  
  const start = Date.now();  
  while (Date.now() - start < 3000) { /* busy wait loop */ }  
  
  pet_info.happiness += 5;  
  pet_info.weight -= 2;  
  triggerPetReaction(imgSleep, "Zzz... That was a nice rest.");  
  checkAndUpdatePetInfoInHtml();  
}
```

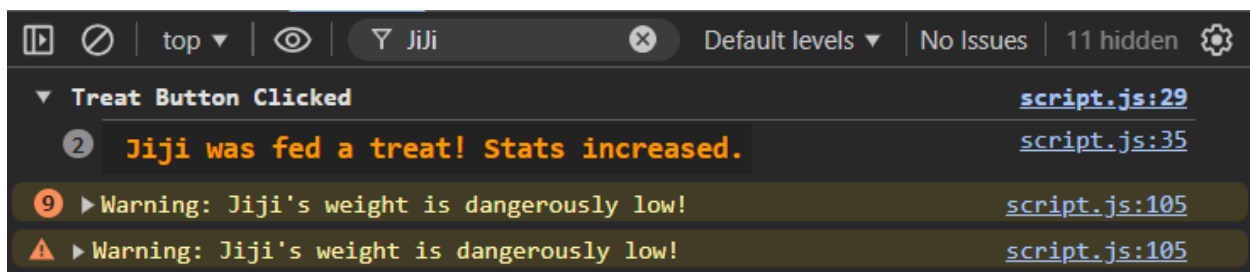
```
[Violation] 'click' handler took 3002ms jquery-2.2.1.min.js:3
```

Filter Messages

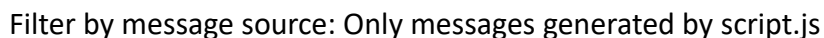
Filter by log level: Only warnings and errors



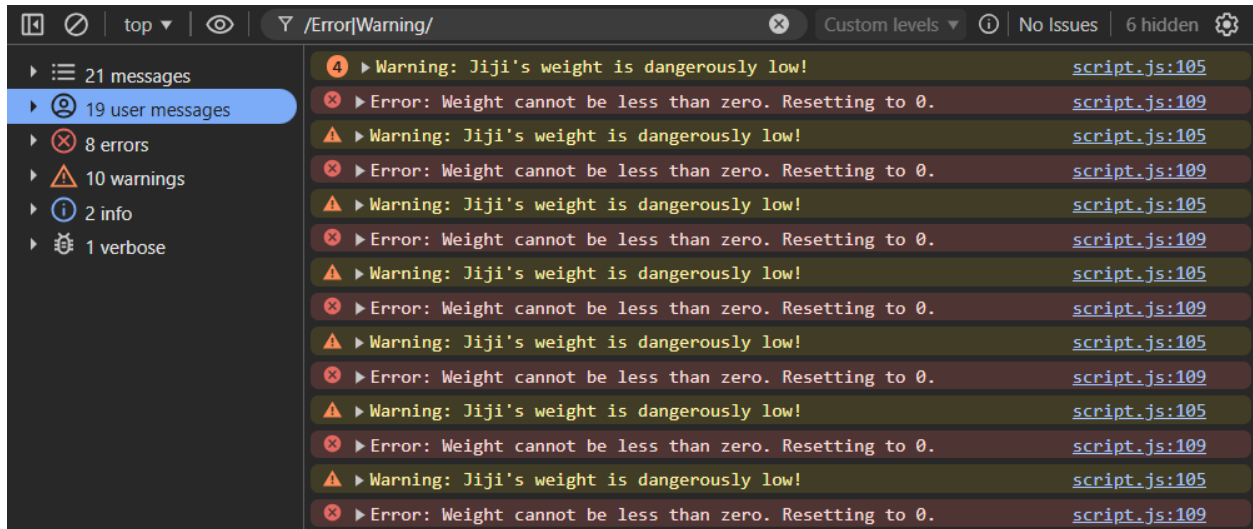
Filter by text: only logs containing text “Jiji”



Filter by regular expression: regex shows only messages containing either “Error” or “Warning”



Filter by user messages:

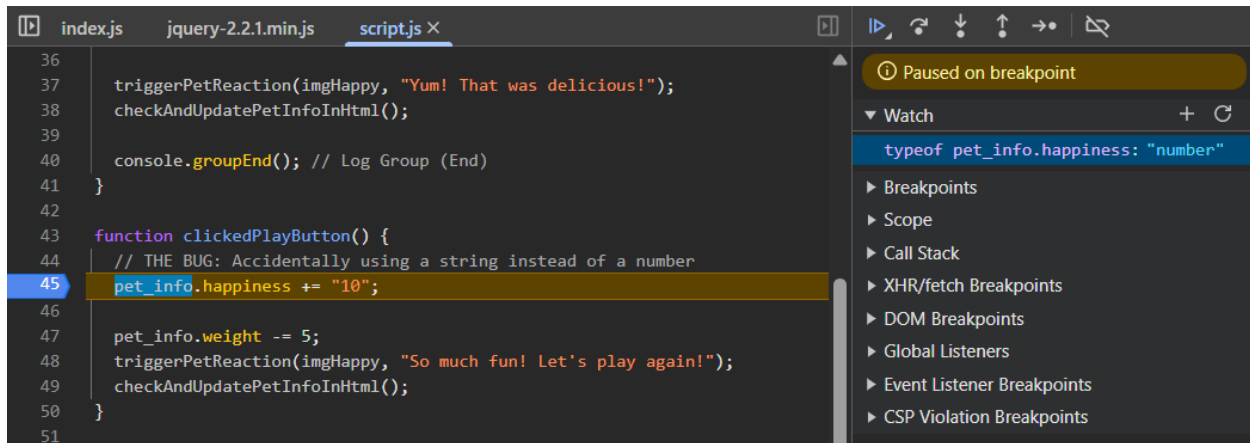


Debugging & Breakpoints

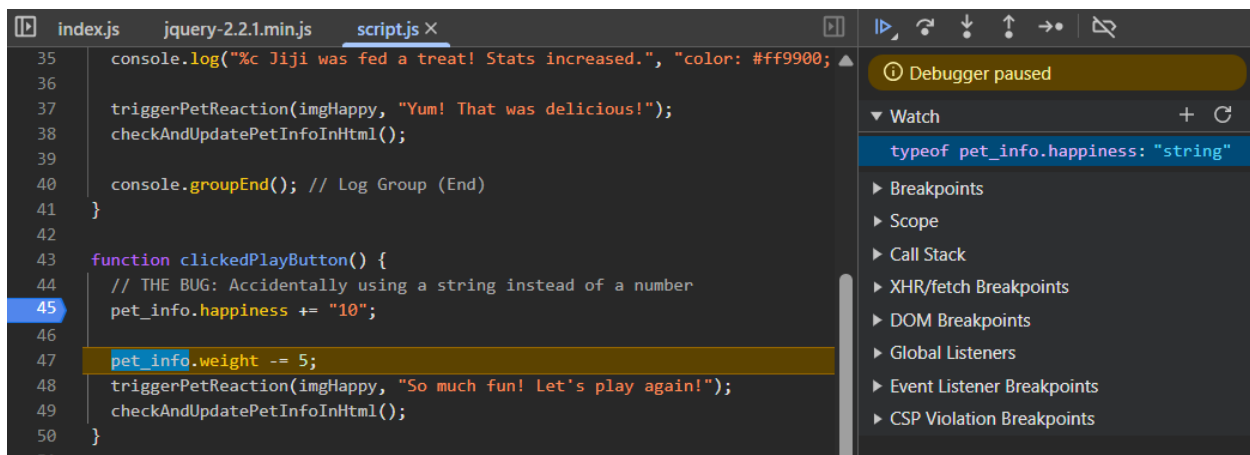
Reproduce a Bug: Inject a bug into the clickedPlayButton function where the number 10 is a string instead of an integer. It now adds 10 happiness as a string ("50" + "10" = "5010") instead of as an integer (50 + 10 = 60).



Set Breakpoint: Bug was introduced by clicking the play button therefore a breakpoint was set on the clickedPlayButton function. The type of happiness is still a number right before execution which is shown on the right panel.



After clicking on step over next function call, type of happiness changes to string. This confirms the source of the bug to be at line 45.



Fix applied by removing quotation marks around 10.

